A FLAT LOCAL HOMOMORPHISM THAT IS UNRAMIFIED ON RESIDUE FIELDS INDUCES AN ISOMORPHISM OF COMPLETIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Flatness of a local homomorphism and the fibre criterion for flatness. used at proof lines 10772-10776. Supplies the flatness hypothesis and the way it is transported to the completion.
- Finiteness and flatness over a Noetherian local ring imply freeness. the author cites Algebra, Lemma algebra-lemma-finite-flat-local at proof lines 10780-10782. Turns flatness into freeness of the completed map.
- Completions of Noetherian local rings and the identification of the residue fields of the source and target. used at proof lines 10782-10787 together with the hypothesis on the residue fields. Provides the rank computation that shows the free module has rank one.

Source reference. Original lemma 0AGX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13210–13210. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0394: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4259–4259. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AGW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13209–13209. Navigation pointer to a statement the

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Source reference. Original lemma 00NZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 877–877. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7060–7060. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a flat local homomorphism of Noetherian local rings such that $\mathfrak{m}_A B = \mathfrak{m}_B$ and $\kappa(\mathfrak{m}_A) = \kappa(\mathfrak{m}_B)$. Then $A \to B$ induces an isomorphism $A^\wedge \to B^\wedge$ of completions.*

Proof. By Algebra, Lemma 0394 we see that $B^\wedge$ is the $\mathfrak{m}_A$-adic completion of B and that $A^\wedge \to B^\wedge$ is finite. Since $A \to B$ is flat we have $\mathrm{Tor}_1^A(B, \kappa(\mathfrak{m}_A)) = 0$. Hence we see that $B^\wedge$ is flat over $A^\wedge$ by Lemma 0AGW. Thus $B^\wedge$ is a free $A^\wedge$-module by Algebra, Lemma 00NZ. Since $A^\wedge \to B^\wedge$ induces an isomorphism $\kappa(\mathfrak{m}_A) = A^\wedge/\mathfrak{m}_A A^\wedge \to B^\wedge/\mathfrak{m}_A B^\wedge = B^\wedge/\mathfrak{m}_B B^\wedge = \kappa(\mathfrak{m}_B)$ by our assumptions (and Algebra, Lemma 05GG), we see that $B^\wedge$ is free of rank 1. Thus $A^\wedge \to B^\wedge$ is an isomorphism. □